

Mohammad Ninad Mahmud Nobo

Bangladesh University of Engineering and Technology (BUET) | CSE

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SUMMARY

CSE undergraduate at BUET focused on building intelligent systems at the intersection of large language models, software automation, and applied machine learning systems. Hands-on experience designing end-to-end AI-driven systems across mobile and backend platforms, with strong emphasis on reliability, scalability, and real-world deployment. Actively engaged in research on LLM-based software testing and medical AI, including dataset creation, evaluation pipelines, and trust-aware decision systems.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

2022 – Present

Bachelor of Science in Computer Science and Engineering (CSE)

CGPA: 3.61 / 4.00

Relevant Coursework: Data Structures and Algorithms, Digital Logic Design, Database Systems, Operating Systems, Computer Architecture, Computer Networks, Computer Graphics, Algorithm Engineering, Artificial Intelligence, Machine Learning

Research Interests: LLM Systems, Software Testing Automation, Applied Machine Learning, Medical AI, Computer Networks

RESEARCH EXPERIENCE

• Undergraduate Thesis: Web Testing Using Large Language Models [GitHub]

Python • LLMs • Web Automation • Prompt Engineering

2025 – Present

- Designing LLM-based pipelines for automated web test generation using prompt-driven and programmatic approaches.
- Creating and curating a dataset of functional descriptions and corresponding test cases for supervised evaluation.
- Investigating coverage improvement and robustness of generated tests.

• MedCAR: Conflict-Aware Medical Reasoning System (Under Review) [GitHub]

Python • Deep Learning • Medical AI

2026

- Built a multi-model system for chest X-ray analysis integrating heterogeneous AI tools.
- Developed a conflict resolution pipeline to reconcile contradictory outputs using semantic reasoning and confidence calibration.
- Designed trust-aware decision framework with uncertainty-based abstention for reliable predictions.

• Bengali-Loop: Community Benchmarks for Long-Form Bangla ASR and Speaker Diarization [arXiv: 2602.14291]

Python • Speech Processing • Machine Learning

2026

- Contributed to building a large-scale benchmark dataset for long-form Bangla ASR and speaker diarization in a multi-institution collaboration, developing data collection and annotation pipelines for real-world speech data.
- Designed subtitle extraction and annotation workflows enabling evaluation with WER and DER.

PROJECTS

• MindTrace: AI-Powered Dementia Care Application [GitHub | Feature | Infrastructure]

Kotlin • Spring Boot • Spring AI • PostgreSQL • Redis • Firebase • Docker • Azure

2025

- Engineered an AI-driven mobile-backend system for dementia care with an Android frontend and scalable Spring Boot architecture, enabling real-time caregiver workflows and seamless system integration.
- Designed accessibility-first UI/UX for cognitively impaired users, focusing on clarity, simplified interactions, and usability to enhance engagement and reduce cognitive load.

• Gemma VetCare: AI-Assisted Farming Support System [GitHub | Feature]

Kotlin • Android Studio • Spring Boot • Spring AI • MongoDB

2025

- Built an AI-assisted Android application for veterinary decision support, integrating backend services to deliver real-time recommendations for diagnosis and treatment workflows.
- Designed RESTful APIs optimized for low-connectivity environments with efficient request handling and streamlined data flow.

• Compiler Construction (Flex, Bison, 8086 Codegen) [GitHub]

C++17 • Flex • Bison/Yacc • Linux

2024

- Designed and implemented a full compiler pipeline for a C-like language, encompassing lexical analysis, parsing, semantic analysis, and intermediate code generation.
- Generated optimized 8086 assembly code with symbol table management and basic optimizations.

• Computer Graphics Pipeline Implementation [GitHub]

C++17 • OpenGL/GLUT • Rasterization • Ray Tracing

2025

- Implemented a complete graphics pipeline including transformations, clipping, rasterization with Z-buffering, and ray tracing.
- Developed OpenGL-based rendering demos with camera systems supporting 6-DOF control and real-time interaction.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, Kotlin, JavaScript, SQL

Backend & APIs: Spring Boot, Spring AI, Node.js, Express.js, REST APIs, Secure API Design

AI/ML: PyTorch, TensorFlow, Scikit-learn, LLM Applications

Databases: PostgreSQL, MongoDB, Redis

DevOps & Infrastructure: Linux, Git, Docker, Docker Compose, GitHub Actions, CI/CD Pipelines, Firebase

ADDITIONAL INFORMATION

Spoken Languages: Bangla (Native), English (Fluent), Hindi & Urdu (Conversational), Arabic (Reading)

Programming Profiles: NeetCode: mninadmnobo LeetCode: mninadmnobo

Research Profiles: ORCID: 0009-0006-2781-6693 ResearchGate: Mohammad Ninad Mahmud Nobo